

Week7 Assignment 9

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Problem 1 Description

Consider the closed-loop response for IMC when the model is not perfect. Show that there is no offset for a setpoint change for the following process model and actual process transfer functions. Also, find the minimum value of λ that assures closed-loop stability for this system.

$$\hat{g}_p(s) = \frac{2}{5s+1}$$

$$g_p(s) = \frac{1.5(-s+1)}{(s+1)(4s+1)}$$

Solution

IMC Controller Design

The Internal Model Control (IMC) controller has the form:

$$q(s) = \frac{\hat{g}_p^{-1}(s)}{f(s)}$$

Where:

$$\hat{g}_p^{-1}(s) = \frac{5s+1}{2}, \quad f(s) = \frac{1}{\lambda s+1}$$

Therefore, the controller becomes:

$$q(s) = \frac{5s+1}{2} \cdot \frac{1}{\lambda s+1} = \frac{5s+1}{2(\lambda s+1)}$$

Closed-Loop Transfer Function

The closed-loop transfer function for setpoint tracking is:

$$T(s) = \frac{g_p(s)q(s)}{1 + q(s)(g_p(s) - \hat{g}_p(s))} = \frac{\frac{1.5(-s+1)}{(s+1)(4s+1)} \cdot \frac{5s+1}{2(\lambda s+1)}}{1 + \frac{5s+1}{2(\lambda s+1)} \cdot \left(\frac{1.5(-s+1)}{(s+1)(4s+1)} - \frac{2}{5s+1} \right)}$$

Steady-State Offset Analysis

Consider a unit step input $r(s) = \frac{1}{s}$. The output is:

$$Y(s) = T(s) \cdot \frac{1}{s}$$

By the Final Value Theorem:

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} T(s)$$

Substituting $s = 0$ into $T(s)$:

$$T(0) = \frac{\frac{1.5(1)}{(1)(1)} \cdot \frac{1}{2(1)}}{1 + \frac{1}{2(1)} \cdot \left(\frac{1.5(1)}{(1)(1)} - \frac{2}{1}\right)} = \frac{\frac{1.5}{2}}{1 + \frac{1}{2} \cdot (-0.5)} = \frac{0.75}{0.75} = 1$$

Thus, a steady-state offset exists. Specifically:

$$\text{Offset} = 1 - T(0) = 0$$

The above proves that there is no offset!

Stability Analysis

Consider the denominator of the closed-loop transfer function:

$$1 + \frac{5s + 1}{2(\lambda s + 1)} \cdot \left(\frac{1.5(-s + 1)}{(s + 1)(4s + 1)} - \frac{2}{5s + 1} \right) = 0$$

$$4\lambda s^3 + (5\lambda - 3.75)s^2 + (\lambda + 3)s + 0.75 = 0$$

$$\begin{array}{c|cc} s^3 & 4\lambda & \lambda + 3 \\ s^2 & 5\lambda - 3.75 & 0.75 \\ s^1 & 5\lambda^2 + 8.25\lambda - 11.25 & 0 \\ s^0 & 0.75 & 0 \end{array}$$

$$\lambda > 0 \quad \lambda > 0.75 \quad \lambda > \frac{-33 + 3\sqrt{521}}{40} \approx 0.8869$$

Therefore, the minimum filter time constant to ensure closed-loop stability is:

$$\lambda > \left(\frac{-33 + 3\sqrt{521}}{40} \right) \approx 0.8869$$

Problem 2 Description

A stack gas scrubber has the following relationship between the fresh feedwater flow rate and the SO₂ concentration in the water leaving the scrubber:

$$y(s) = \frac{-2(6s + 1)}{(10s + 1)(3s + 1)} u(s)$$

Use the IMC-based PID procedure to find the PID controller (plus first-order lag) for this process.

Hint: Let the IMC $q(s)$ be semiproper (order of denominator = order of numerator). Which controller tuning parameters does λ affect?

Solution

Step 1: Identify the process model $g_p(s)$

$$g_p(s) = \frac{-2(6s + 1)}{(10s + 1)(3s + 1)}$$

This is a second-order system with a negative steady-state gain (non-minimum phase behavior). Let us factor this expression to make it easier to work with.

$$g_p(s) = \frac{-2(6s + 1)}{(30s^2 + 13s + 1)}$$

Step 2: IMC Structure

We design an internal model control filter:

$$q(s) = \frac{g_p^{-1}(s)}{f(s)}$$

where $f(s)$ is the low-pass filter:

$$f(s) = \frac{1}{\lambda s + 1}$$

We choose $f(s)$ to make $q(s)$ semiproper, i.e., the same number of poles as zeros. Since $g_p(s)$ is proper (2 poles, 1 zero), we add one pole in the filter to make the IMC controller semiproper.

Now, the inverse of $g_p(s)$ is:

$$g_p^{-1}(s) = \frac{(10s + 1)(3s + 1)}{-2(6s + 1)}$$

Thus, the IMC controller becomes:

$$q(s) = \frac{(10s + 1)(3s + 1)}{-2(6s + 1)(\lambda s + 1)}$$

Step 3: Convert to standard feedback form

We convert the IMC controller to a standard feedback controller:

$$c(s) = \frac{q(s)}{1 - g_p(s)q(s)}$$

With algebraic simplification (or from the standard IMC-to-PID derivation), the resulting controller $c(s)$ is equivalent to a PID controller with a first-order lag:

$$c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right) \cdot \frac{1}{\tau_f s + 1}$$

Step 4: Use IMC-PID correlations

$$c(s) = \frac{q(s)}{1 - g_p(s)q(s)} = \frac{(10s + 1)(3s + 1)}{(-2)(6s + 1)(\lambda s)} = \frac{13}{-2\lambda} \left(1 + \frac{1}{13s} + \frac{30}{13}s \right) \cdot \frac{1}{6s + 1}$$

$$K_c = \frac{13}{-2\lambda}$$

$$\tau_I = 10 + 3 = 13$$

$$\tau_D = \frac{10 \cdot 3}{10 + 3} = \frac{30}{13}$$

$$\tau_f = 6$$

Final PID Controller with First-Order Lag

The final PID controller is:

$$c(s) = \frac{13}{-2\lambda} \left(1 + \frac{1}{13s} + \frac{30}{13}s \right) \cdot \frac{1}{6s + 1}$$

Effect of λ

The parameter λ acts as a tuning knob:

- Larger λ : smoother response, better robustness, but slower performance.
- Smaller λ : faster response, more aggressive control, but reduced robustness.

It directly affects:

- Controller gain K_c